



**EAST TEXAS A&M**  
— UNIVERSITY —

# CSCI 352.61E

## DIGITAL FORENSICS

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# **System Forensics, Investigation, and Response**

## **Lesson 4 Collecting, Seizing, and Protecting Evidence**

# Learning Objective

- Outline the proper approach to collecting, seizing, and protecting evidence.

# Key Concepts

- Proper forensic procedure
- Evidence handling
- Storage formats
- Forensically imaging a drive
- RAID acquisitions



# Proper Procedure Overview

Shut down the computer.

Transport the computer to a secure location.

Prepare the system.

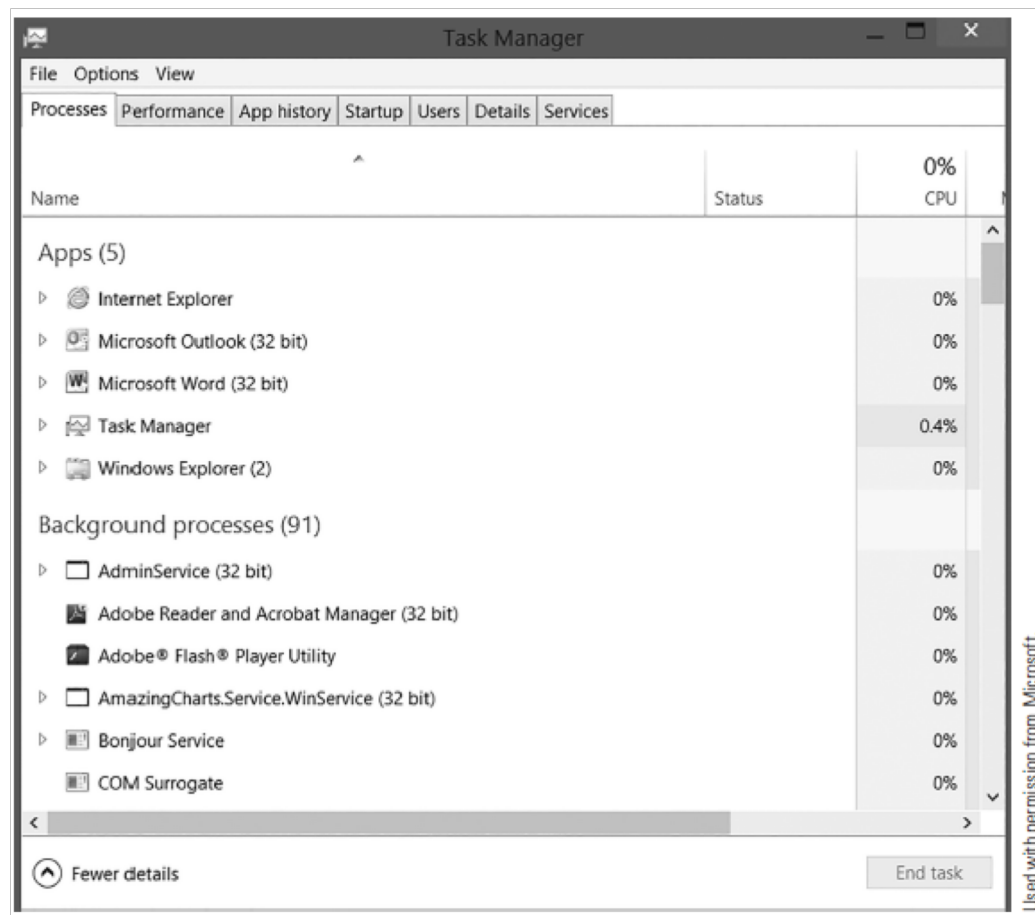
Document the hardware configuration of the system.

Mathematically authenticate data on all storage devices.

# Shutting Down the Computer

- Before you shut the computer down:
  - Check for running processes
    - In Windows, use Task Manager
  - Take a picture of the screen for your records
  - Check for live connections to the system:
    - `netstat`
    - `net sessions`
    - `open files` (critical to run)

# Task Manager Running Processes



# netstat Command

```
C:\Windows\system32\cmd.exe - netstat

Microsoft Windows [Version 6.2.9200]
(c) 2012 Microsoft Corporation. All rights reserved.

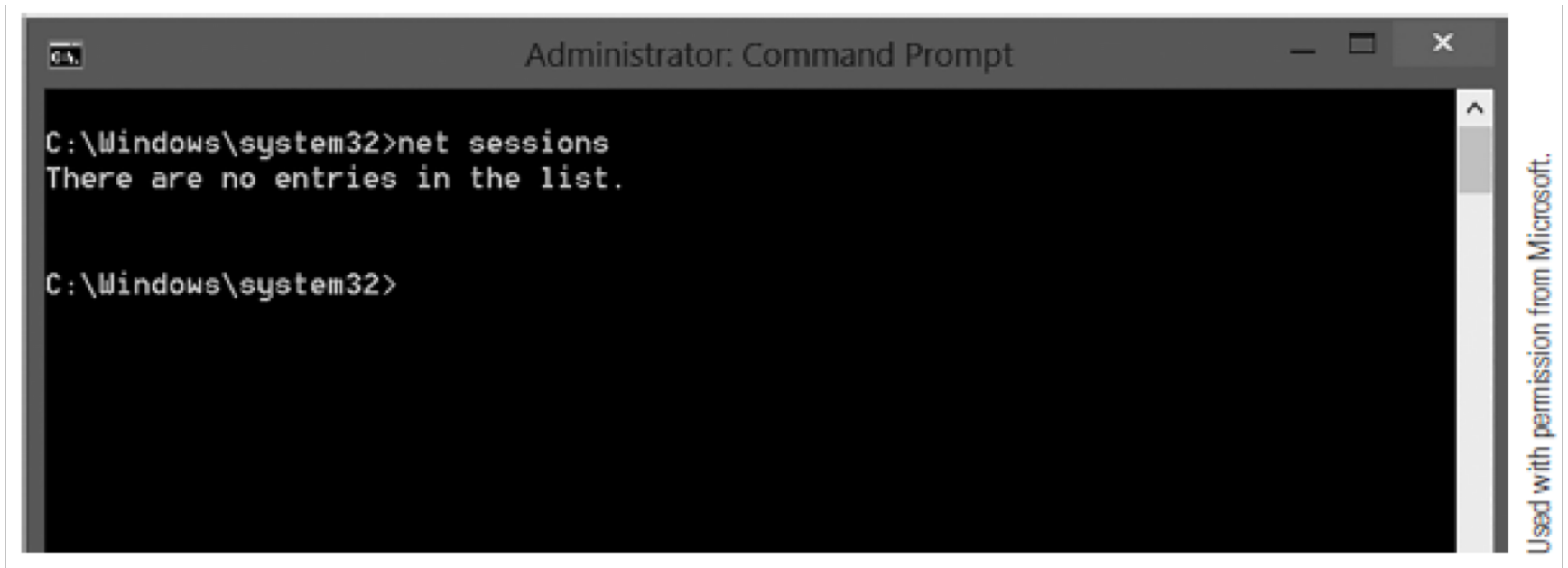
C:\Users\chuckeasttom>netstat

Active Connections

    Proto Local Address          Foreign Address         State
    TCP    127.0.0.1:1030          ChucksLaptop:5354      ESTABLISHED
    TCP    127.0.0.1:1104          ChucksLaptop:27015     ESTABLISHED
    TCP    127.0.0.1:5354          ChucksLaptop:1030      ESTABLISHED
    TCP    127.0.0.1:27015         ChucksLaptop:1104      ESTABLISHED
    TCP    192.168.1.5:1539        server-54-240-170-126:http CLOSE_WAIT
```

Used with permission from Microsoft.

# net sessions Command

A screenshot of a Windows Command Prompt window titled "Administrator: Command Prompt". The window has a standard Windows title bar with minimize, maximize, and close buttons. The command prompt shows the following text:

```
C:\Windows\system32>net sessions
There are no entries in the list.

C:\Windows\system32>
```

The text is displayed in a white monospaced font on a black background. A vertical scrollbar is visible on the right side of the command prompt window. To the right of the command prompt window, there is a vertical text label that reads "Used with permission from Microsoft."

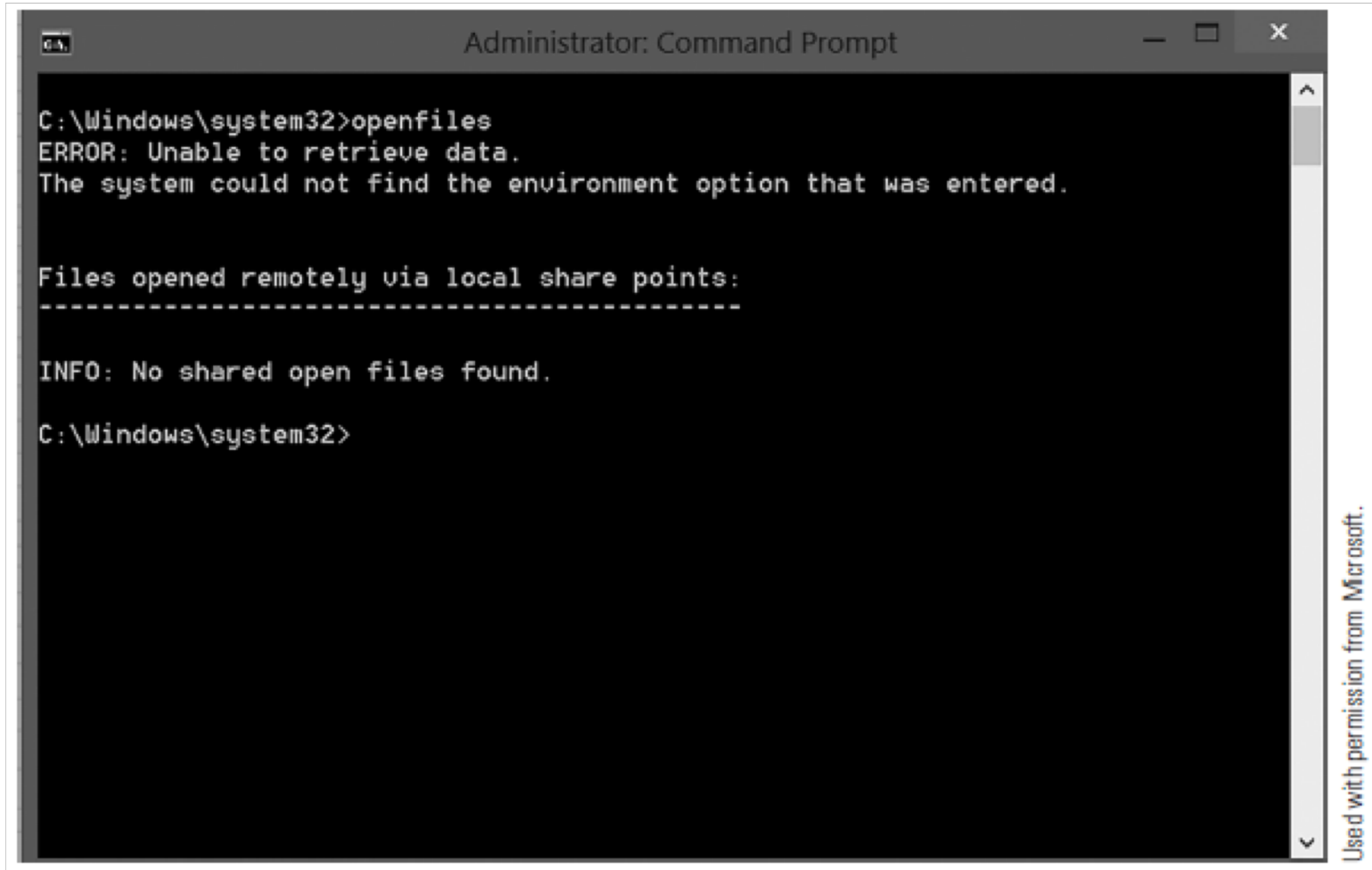
Administrator: Command Prompt

```
C:\Windows\system32>net sessions
There are no entries in the list.

C:\Windows\system32>
```

Used with permission from Microsoft.

# openfiles Command



The screenshot shows a Windows Command Prompt window titled "Administrator: Command Prompt". The command prompt is at the directory "C:\Windows\system32". The user has entered the command "openfiles". The output shows an error: "ERROR: Unable to retrieve data. The system could not find the environment option that was entered." This is followed by a section header "Files opened remotely via local share points:" with a dashed line underneath. Below this, it says "INFO: No shared open files found." The prompt then returns to "C:\Windows\system32>".

```
C:\Windows\system32>openfiles
ERROR: Unable to retrieve data.
The system could not find the environment option that was entered.

Files opened remotely via local share points:
-----

INFO: No shared open files found.

C:\Windows\system32>
```

Used with permission from Microsoft.

# System Memory with OSForensics

Courtesy of PassMark Software.

OSForensics - Moriarity Case

**Memory Viewer**

Refresh Select Window Dump Physical Memory Save Crash Dump

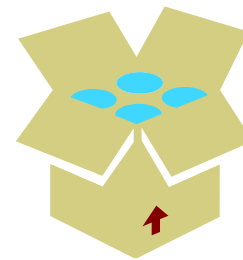
| Process              | PID  |
|----------------------|------|
| nvsvc.exe            | 1240 |
| nvdsync.exe          | 1228 |
| OfficeClickToRun.exe | 2360 |
| osf64.exe            | 9136 |

Process Info Memory Space Memory Layout

| Address Range             | Size  | State  | Protection | Type   | Module              |
|---------------------------|-------|--------|------------|--------|---------------------|
| 0x61611000 - 0x61627FFF   | 92 KB | Commit | XR         | Module | mdnsNSP.dll.text(1) |
| 0x140001000 - 0x1407A4FFF | 8 MB  | Commit | XR         | Module | osf64.exe!.text(1)  |
| 0x1407A5000 - 0x1407A5FFF | 4 KB  | Commit | XRW        | Module | osf64.exe!.text(1)  |
| 0x145CDA000 - 0x145CDBFFF | 8 KB  | Commit | X          | Module | osf64.exe!.text(6)  |
| 0x145CE2000 - 0x14646BFFF | 8 MB  | Commit | XWC        | Module | osf64.exe!.rsrc(8)  |
| 0x14646C000 - 0x14646CFFF | 4 KB  | Commit | XRW        | Module | osf64.exe!.rsrc(8)  |

# Transporting the Computer

- Keep evidence in possession or control at all times
- Document movement of evidence between investigators
- Secure evidence appropriately so it can't be tampered with or corrupted
- Lock in a vehicle
- Drive vehicle directly to lab





# Preparing the System

- For suspect computers:
  - Remove the drive(s)
  - Create an evidence form and/or a chain of custody form
- For mobile devices:
  - Remove SIM card, if necessary
  - Some devices let you dock the phone examine it without removing SIM

# Evidence Form

| Single Evidence Form                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                              |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| <div style="display: flex; justify-content: space-around;"> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> <div style="border: 1px solid black; width: 20px; height: 20px;"></div> </div> <div style="display: flex; justify-content: space-around; font-size: small;"> <span>Case No.</span> <span>Evidence No.</span> </div> | <div style="text-align: right; font-size: small;"> </div>                    |
| PLEASE COMPLETE FORM IN UPPERCASE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                              |
| <b>Section B: Evidence Collection</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                              |
| Date/Time Collected <span style="border: 1px solid black; padding: 2px;">DDMMYY</span> <span style="border: 1px solid black; padding: 2px;">HH:MM</span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Collected by                                                                 |
| Site Address<br><hr/> <hr/> <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                              |
| <b>Section C: Evidence Details</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                              |
| Date/Time Stored <span style="border: 1px solid black; padding: 2px;">DDMMYY</span> <span style="border: 1px solid black; padding: 2px;">HH:MM</span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                              |
| Storage Location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                              |
| Device Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Capacity                                                                     |
| Manufacturer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Model                                                                        |
| Serial No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                              |
| MD5 Sum <span style="border: 1px solid black; display: inline-block; width: 100px; height: 1.2em;"></span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                              |
| SHA-1 Sum <span style="border: 1px solid black; display: inline-block; width: 150px; height: 1.2em;"></span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                              |
| Additional Information...<br><hr/> <hr/> <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                              |
| Note any damage, marks and scratches<br><hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Digital Image Taken <input type="checkbox"/> Yes <input type="checkbox"/> No |
| <b>Section D: Image Details</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                              |
| Date/Time Imaged <span style="border: 1px solid black; padding: 2px;">DDMMYY</span> <span style="border: 1px solid black; padding: 2px;">HH:MM</span>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Imaged by                                                                    |
| Storage Location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                              |
| Image Filename                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Image Size <span style="font-size: x-small;">(inc. unit)</span>              |
| Additional Information...<br><hr/> <hr/> <hr/>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                              |

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# Chain of Custody Form

| CHAIN OF CUSTODY |      |         |                      |
|------------------|------|---------|----------------------|
| Date             | Time | Officer | Locker # or Location |
|                  |      |         |                      |
|                  |      |         |                      |
|                  |      |         |                      |
|                  |      |         |                      |
|                  |      |         |                      |
|                  |      |         |                      |
|                  |      |         |                      |
|                  |      |         |                      |
|                  |      |         |                      |

Received from the McKinney Police Department the item(s) shown on the reverse side of this card. I hereby certify that I am authorized to take possession and that I release the McKinney Police Department from any and all responsibility.

Witness: \_\_\_\_\_ Signature: \_\_\_\_\_

Date: \_\_\_\_\_ Address: \_\_\_\_\_

OWNER: \_\_\_\_\_  
ADDRESS: \_\_\_\_\_



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# Documenting the Hardware Configuration of the System

- Before dismantling the computer:
  - Take pictures of computer from all angles
  - Record BIOS system time and date in chain of custody form
- After restoring power:
  - Eject all removable media and fill out a separate chain of custody form for each

# Mathematically Authenticating Data on All Storage Devices

- After imaging drive, create a hash of the original and the copy
- Compare the hashes
- If they don't match exactly, something was altered
- Document hashing algorithm used and results

# Handling Evidence

- Digital forensics specialist is responsible for finding, preserving, and preparing evidence
- The specialist must:
  - Collect data
  - Document filenames, dates, and times
  - Identify any file, program, and storage anomalies
  - Gather evidence

# Collecting Data

- Three primary types of data that a forensic investigator must collect, in this order:
  1. Volatile data
  2. Temporary data
  3. Persistent data

# Documenting Filenames, Dates, and Times

- Filenames, creation dates, and last modified dates and times can be relevant as evidence
- Catalog all allocated and “erased” files
- Sort files based on filename, file size, file content, creation date, and last modified date and time
- Sorting provides a timeline of computer usage



# Identifying File, Program, and Storage Anomalies

- Text search programs can't identify text data stored in binary format
  - They require manual evaluation
- Evaluate hidden partitions for evidence and document their existence
- In Windows, also evaluate files in the Recycle Bin

# Evidence-Gathering Measures

Avoid changing evidence

Determine when evidence was created

Search throughout a device

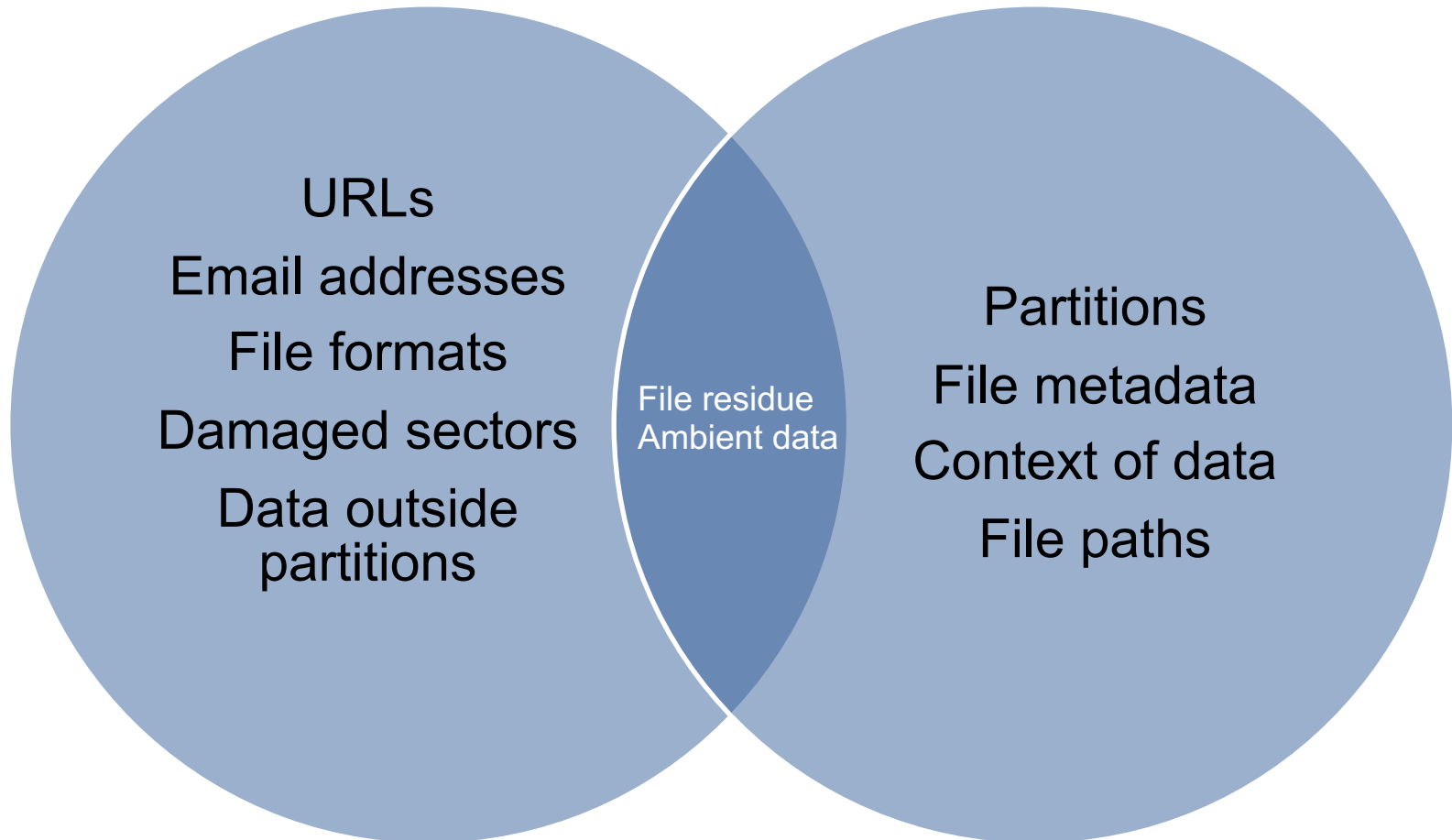
Determine information about encrypted and steganized files

Present evidence well

# Physical and Logical Analysis

## Physical Analysis

## Logical Analysis



# Creating a Timeline

- To reconstruct events that led to corruption of a system, create a timeline
- Challenges with computers:
  - Clock drift
  - Delayed reporting
  - Different time zones
- Never change clock on a suspect system
- Record clock drift and time zone in use

# Storage Formats

Magnetic media

Solid-state drives

Digital audio tape (DAT) drives

# Storage Formats (cont.)

Digital linear tape (DLT) and super DLT

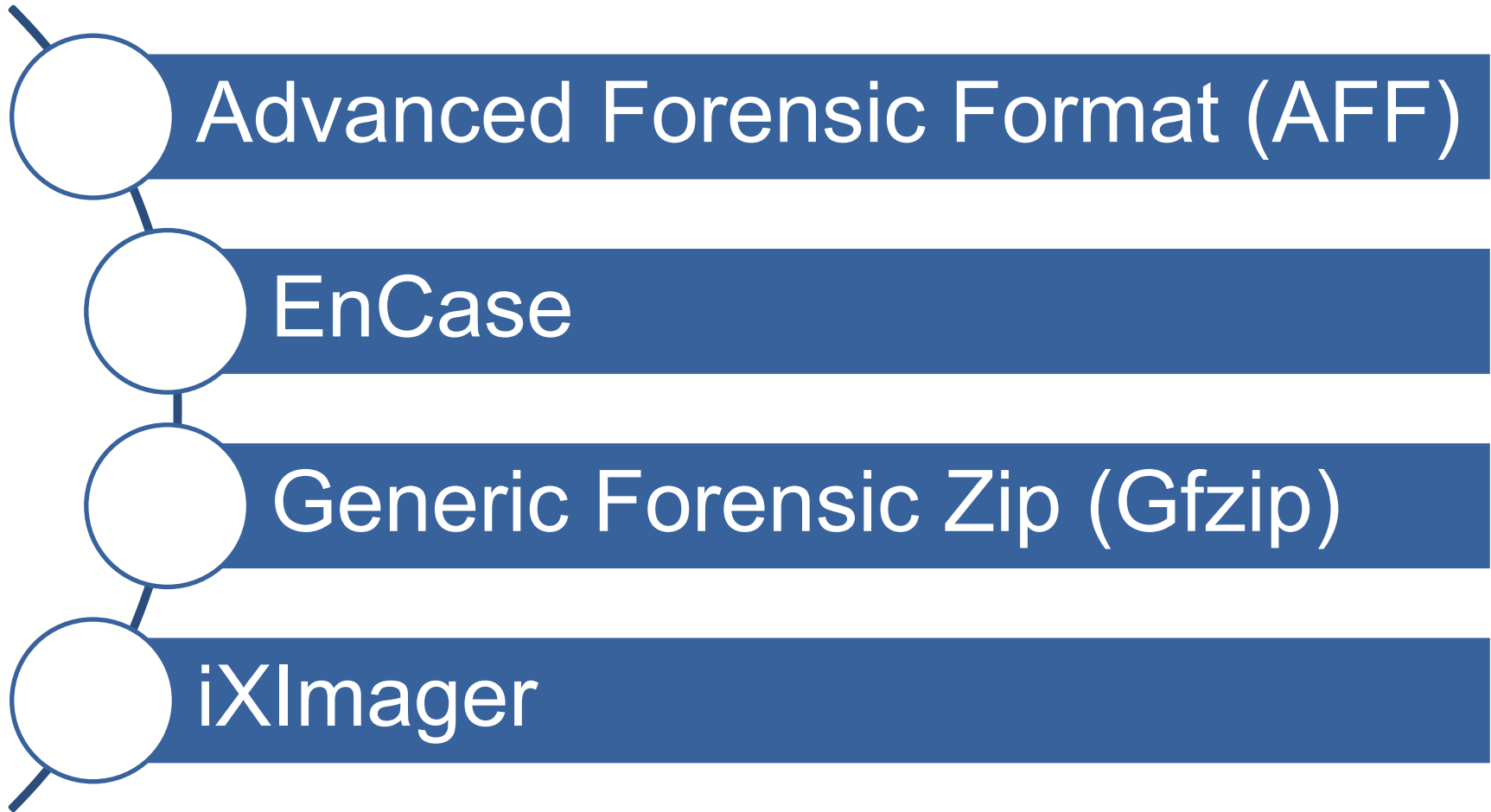
Optical media

Universal serial bus (USB)

# Magnetic and Solid State Drives

- Host protected area (HPA) or vendor-specific drive space
- Master boot record (MBR) where empty drive sectors remain
- Volume slack
- Unallocated space
- Good blocks marked as bad
- File slack

# File Formats





# Forensic Imaging

- An image is an exact bit-by-bit copy of a disk
- Used for evidence collection without altering original

# Imaging with *dd* and *netcat*

## 1. Forensically wipe the drive:

```
dd if=/dev/zero of=/dev/hdb1 bs=2048
```

## 2. Use *netcat* to set up the forensic server to listen:

```
# nc -l -p 8888 > evidence.dd
```

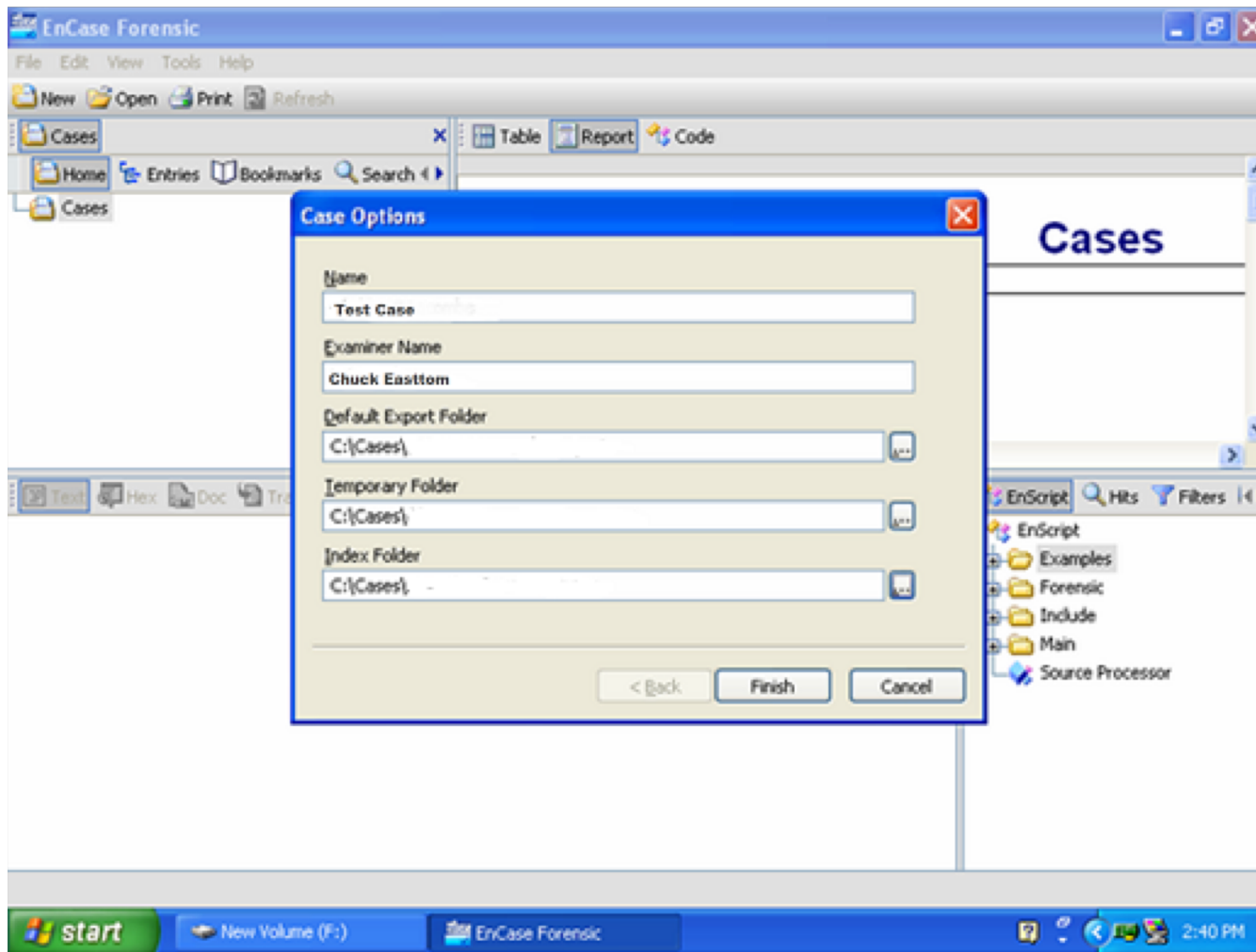
## 3. Use the *dd* command to read the first partition:

```
# dd if=/dev/hda1 | nc 192.168.0.2  
8888 -w 3
```

# Imaging with EnCase

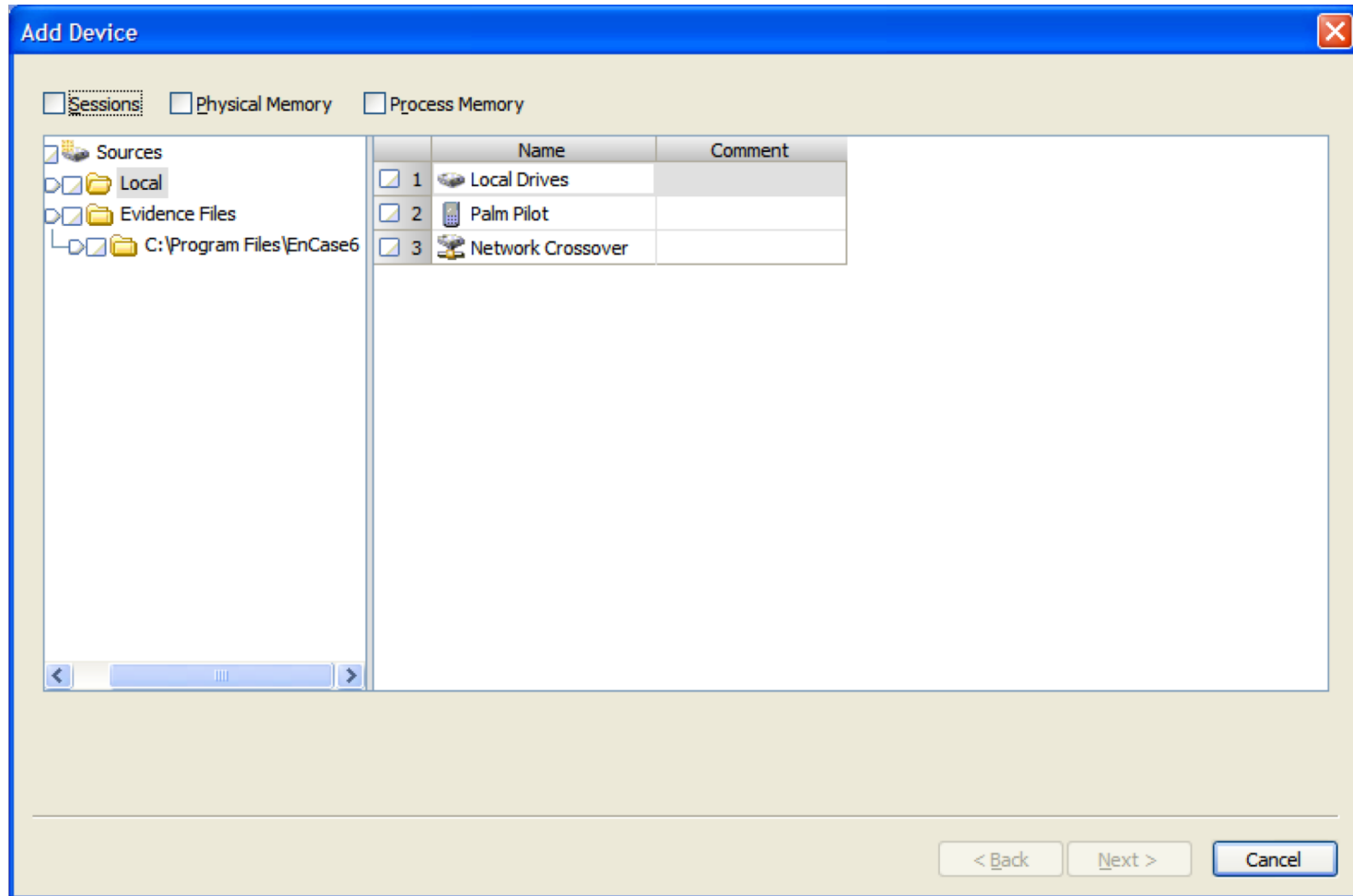
- Makes bit-level images and then mounts them for analysis
- Preview mode allows investigator to use a null modem cable or Ethernet connection to view data on the subject machine safely
- Doesn't alter evidence

# Imaging with EnCase



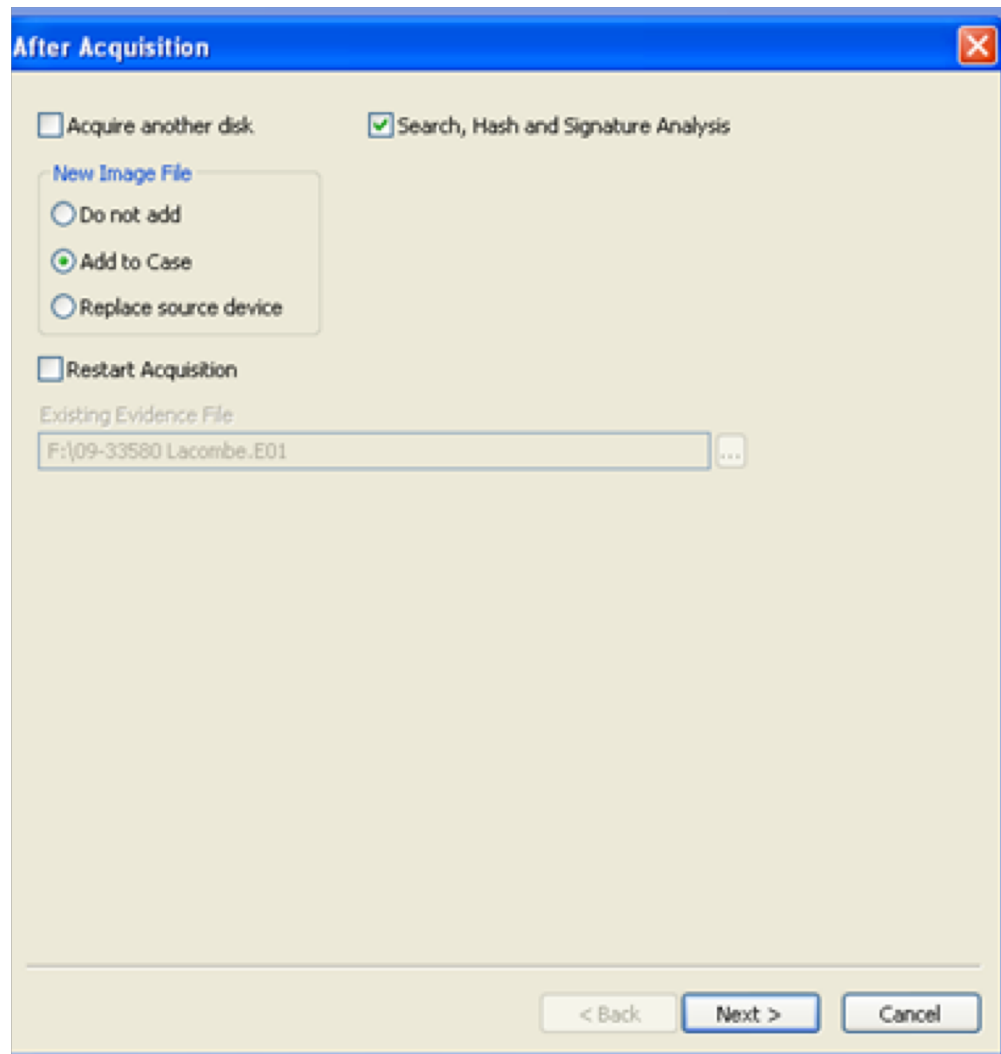
Courtesy of Guidance Software, Inc.

# Imaging with EnCase



Courtesy of Guidance Software, Inc.

# Imaging EnCase

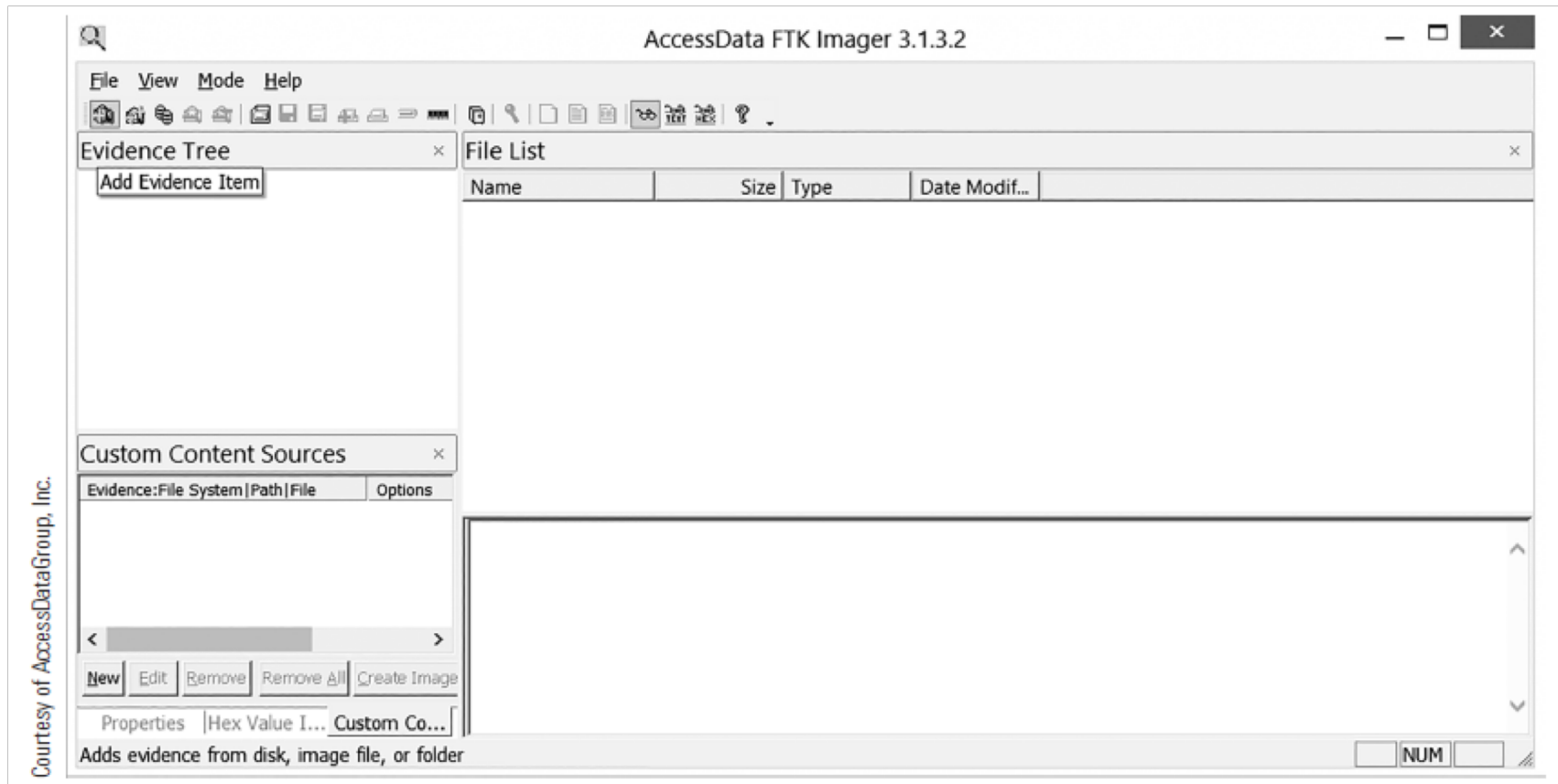


Courtesy of Guidance Software, Inc.

# Imaging with Forensic Toolkit (FTK)

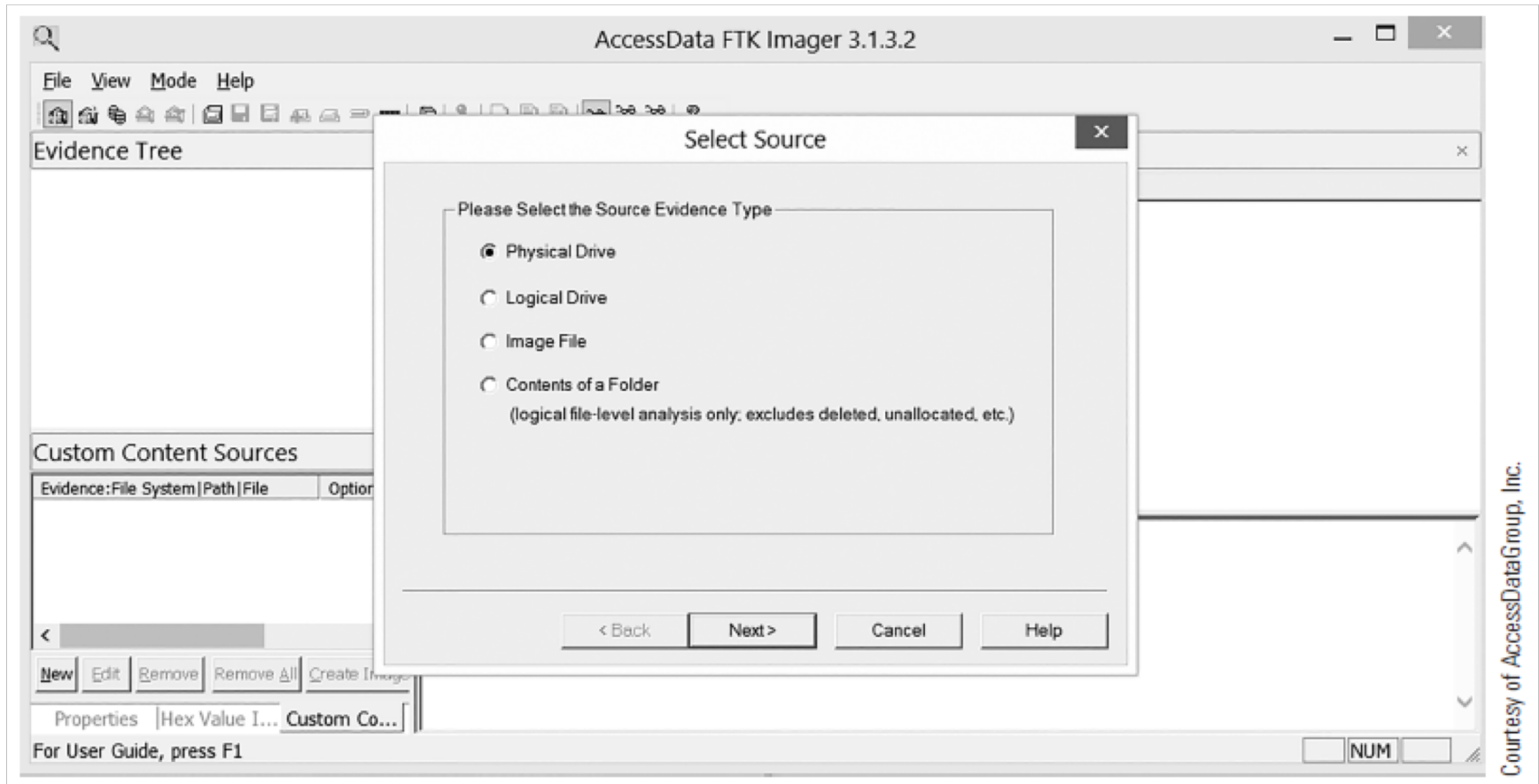
- Takes snapshot of entire disk, makes bit-level copy for analysis
- Inexpensive, easy to use, good all-in-one forensic tool
- Offers Registry viewing, in-depth logging, standalone disk imaging, direct email and zip file analysis

# Imaging with FTK

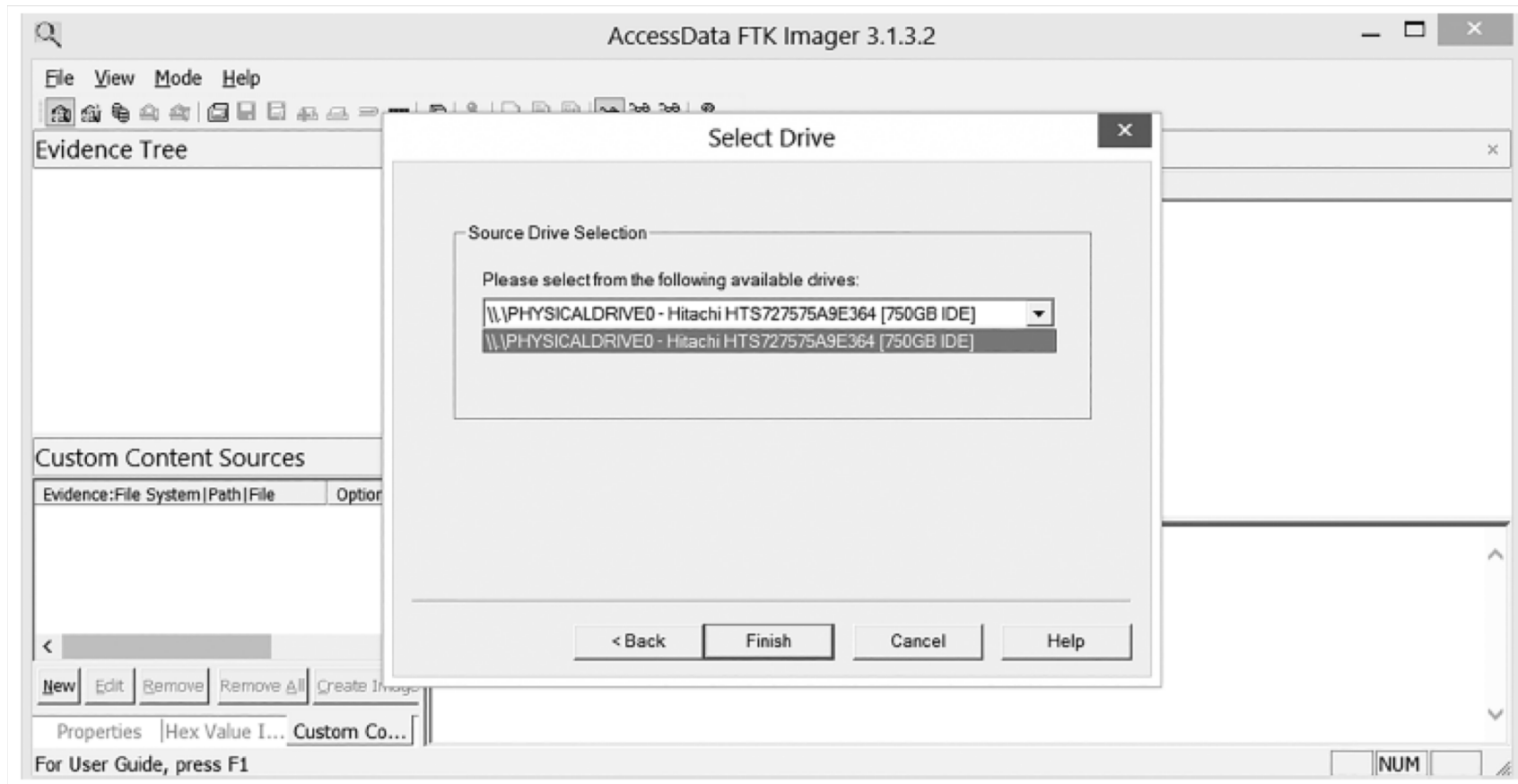




# Imaging with FTK



# Imaging with FTK



Courtesy of AccessData Group, Inc.

# Imaging with OSForensics

- Allows you to mount images created with other tools
- Also allows you to create an image

# Imaging with OSForensics

**Drive Imaging**

Create Image | Restore Image | Hidden Areas - HPA/DCO | RAID Rebuild

Source Disk: \\.\PhysicalDrive0: Partition 3, C: (931.00GB NTFS/H)

Target Image File:  ...

Description:

Location/Place:

☒ Verify Image File After Completion

☒ Disable Shadow Copy

☐ Attach Image Metadata File to Case on Completion

Status:

Copy Method:

Data Read:  Disk Size:

Speed:  Unreadable Data:

Create Image

Courtesy of PassMark Software.

# RAID Acquisitions

- RAID 0, 1, 3, 4, 5, 6, 1+0
- Can acquire RAID 1 disks separately
- RAID 0, 3, 4, 5, and 6 use data striping across multiple disks
  - Make a forensic image of the entire RAID array

# Redundant Array of Independent Disks (RAID)

## RAID 0

- Distributes data across multiple disks
- Improved data retrieval speed

## RAID 1

- Mirrors disks
- Creates an identical copy of a drive

## RAID 3 or 4

- Combines three or more disks for fault tolerance
- Includes parity

# Redundant Array of Independent Disks (RAID)

## RAID 5

- Combines three or more disks
- Similar to RAID 3 but no parity

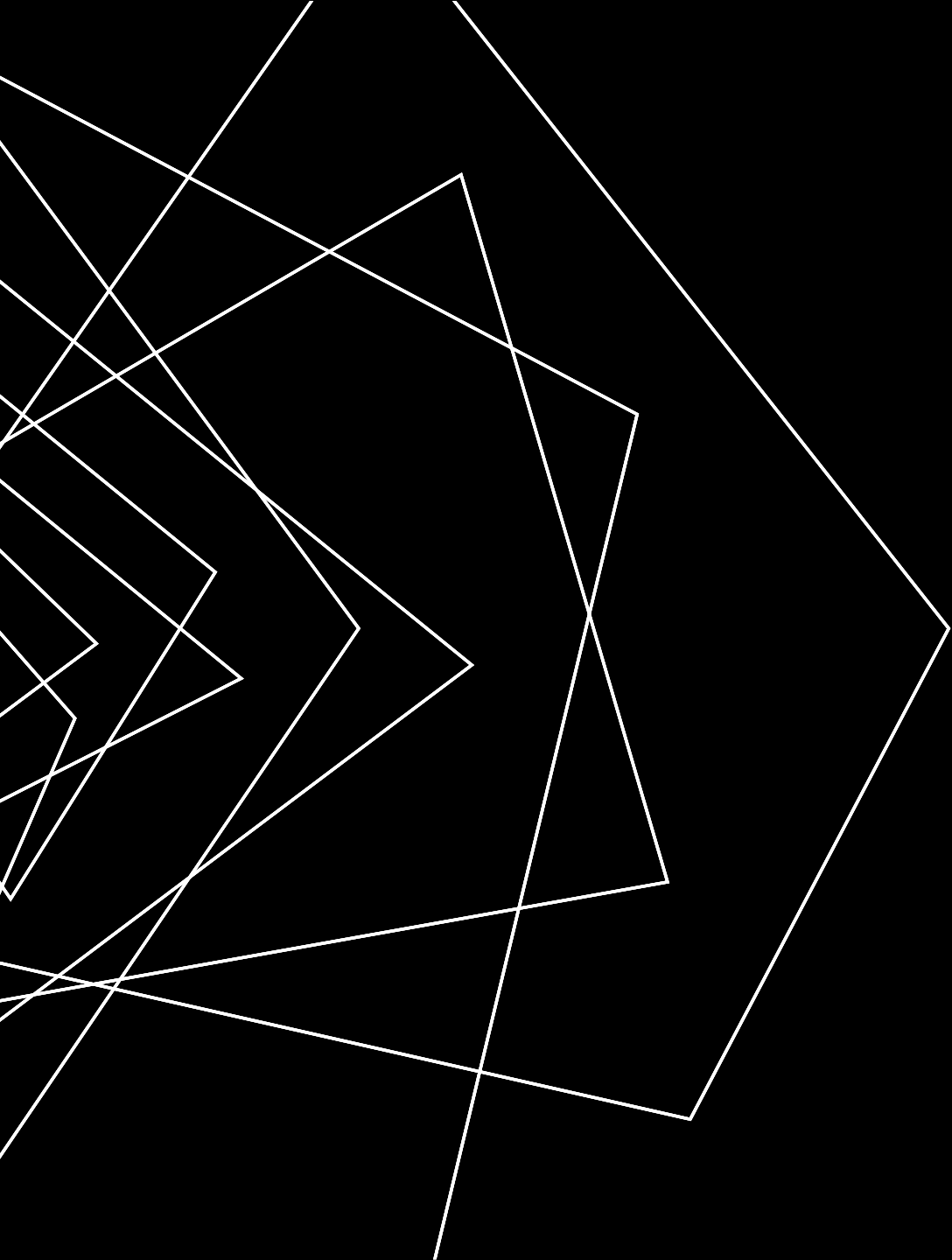
## RAID 6

- Combines four or more disks
- Protects against loss of data on any two disks

# Summary

- Proper forensic procedure
- Evidence handling
- Storage formats
- Forensically imaging a drive
- RAID acquisitions





# THANK YOU

Yau Ka Cheung  
Yauka.Cheung@etamu.edu  
ACB1 Room 335